

2.4 File Attributes : File Permissions listing file, permissions, chmod Command

Various File access Permissions Commands:

The command *chmod* (short for charge made) allows to decide who can and can not read, write and execute a file.

To assign these types of permissions, the following three symbols are used.

Table 1. Commands and their syntax

Access type	Symbol	What it means
Read	r	Allows displaying, coping and compiling a file
Write	w	Allows editing and deleting of a file
Execute	x	Allows execution of a file

To specify then user to whom you are grating these type of permissions, use these three symbols.

Symbols	Description
u	The user or owner of file and directories
g	Members of the group o which user belongs
o	All other system users (users who are neither u or g)

Normally when a file is created, it is created with read, write and execute

Permissions to user and only read and execute permission for group members and other system users

Write permission is not given normally for these (group and other user) two categories. If it is required, the user can permissions by system commands.

HOW TO DETERMINE EXISTING PERMISSIONS:

The user can determine what permissions are currently in effect on a file or a directory by using command that produces a long listing of a directory's

Contents, that is **ls -l**

Example:

Assume the user is logged in under name 'vijay'

\$ ls -l <Enter>

The system displays the following.

Total 3

drwx - -x - -x vijay project 50 june 2 15:30 dir1

-rwx r - x r-x vijay project 1120 june 1 09:50 prog1

-rwx r - -r - - vijay project 2220 june 6 10:40 sample1

Permissions for the prog1 and sample1 files and directory dir1 are shown above

Permissions and their explanations for the above example are shown in the below table.

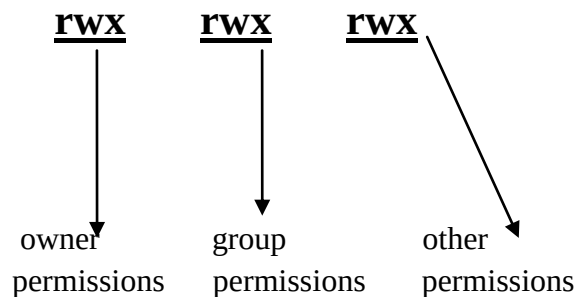
Permissions	Files/directory	Remarks
-rwx r - x r - x	Prog1 (file)	User/owner has got all the three permissions Group and Others have got only read and execute permissions
-rwx r - - r - -	Sample1 (file)	Sample1 defines write and execute permissions For group members and others
drwx r - - r - -	dir1 (directory)	dir1 directory denies permission for write and execute to group and others

Initial character, which describes the file or directory

‘-‘ (dash) symbolizes an ordinary file

‘d’ symbolizes a directory

There are nine possible permissions which can be given or refused in combination. They are written as a string of nine characters which are shown below:



e.g. rwx r-x r-x

In this, example user(owner) has read(r),write(w),execute(x) permissions and the members of group and other system users have read(r)and execute(x)

Permissions only.

HOW TO CHANGE EXISTING PERMISSIONS:

After you have determined what permission are to effect, you can change them by executing ‘*chmod*’ command in following format.

They are two ways of using ‘*chmod*’ command.

They are :

- 1) Symbolic method
- 2) Octal method

1) Symbolic Method

Syntax of symbolic method is as follows

chmod who + permissions files(s) <Enter>

where:

chmod - name of program (command)

who - one of three user groups (u,g,o or a)
 u = user/owner , g = group,
 o = others, a = all

‘+’ - means grant permissions
 ‘-’ - means deny permissions

permissions - any combination of three authorizations (r,w and x)
 files(s) - file or directory names(s) listed.

e.g. \$ chmod u+r prog1

In this, example, read permission has been added to the owner.

e.g. \$ chmod u-w <Enter>

In this, example owner has been denied write permission.

Note: If you omit scope (“u”, ”g” or “o”) the permission given apply to all the
 Three groups.

2) Octal Method:

This format requires to specify permissions using three octal numbers, ranging from 0 to 7. The octal code can be one to three digits long. The first, second, third digits set the “user”, “group” and “other” user permissions respectively. The integer used at each position dictates the permission for the file.

owner	group	other
r w x	r w x	r w x
(4) (2) (1)	(4) (2) (1)	(4) (2) (1)

The meaning of these integers is as follows

Integer	Description
0	is no permission
1	is execute permission
2	is write permission; 3- is write and execute permission
4	is read permission
5	is execute and read permission
6	is write and read permission
7	is read , write and execute permission

e.g. \$ chmod 750 sample1

(111 101 000) i.e. (r w x r- x - - -) for User (Owner), Group & Other member

This example changes access permissions for the file sample1 in the current working directory, the '7' gives permission for the owner of the file to read , write and execute file, the '5' gives permission for group members that file and '0' denies permission for users to read, write or execute the file "sample".